Head and Neck Embryology

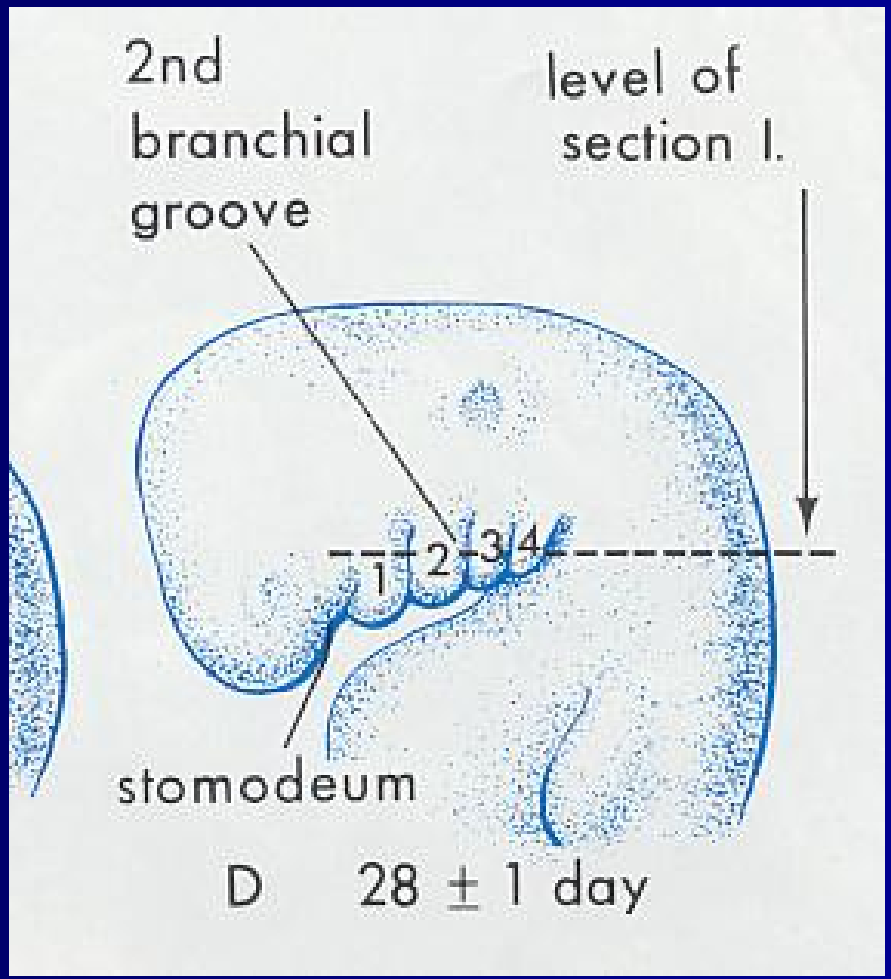
Head & Neck Embryology

- Branchial Apparatus
- Thyroid Gland
- Tongue
- Development of the face
  - Nose
  - Palate
- Ear

Branchial Apparatus

- 4 arches are well developed by 4th week of gestation
- 5th and 6th arches are still rudimentary
- Development takes place over weeks 4 to 7
- Contribute mostly to neck development but the first arch contributes to facial development

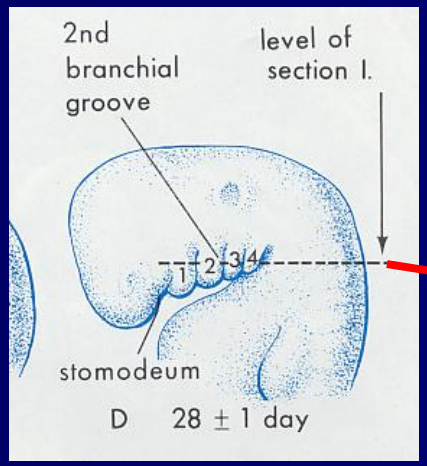
Branchial Apparatus

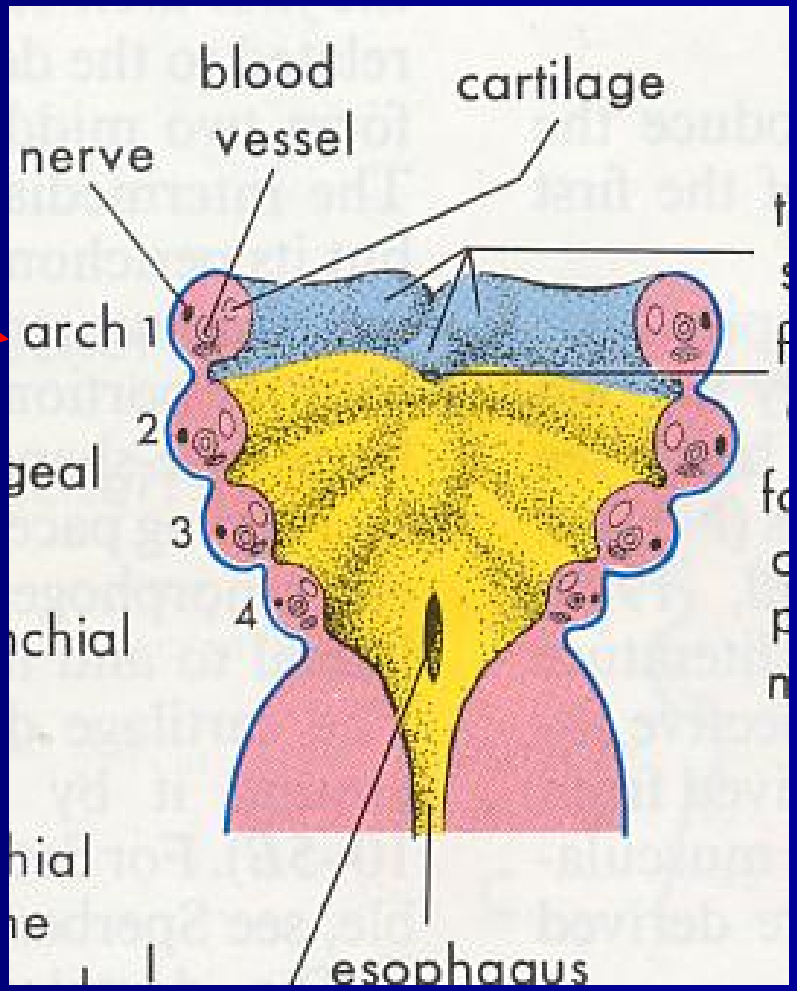


Branchial Apparatus

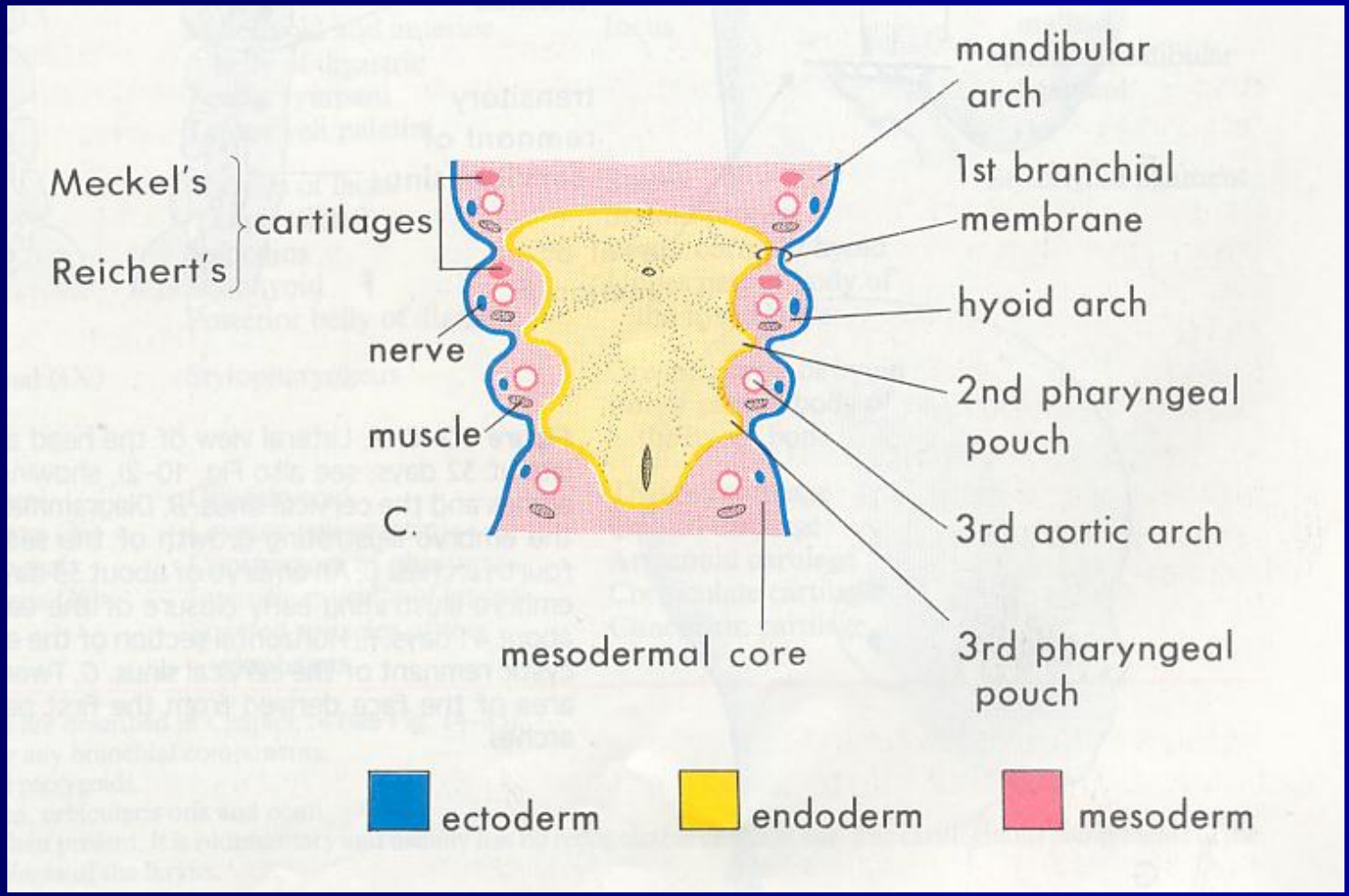
- Arches
  - mesenchymal tissue surrounded by ectoderm and endoderm
- Clefts (or grooves)
  - Separate adjacent arches along ectodermal surface
- Pouches
  - Outpouching of endoderm from foregut
  - Penetrate adjacent mesenchyme

Branchial Apparatus





Branchial Apparatus



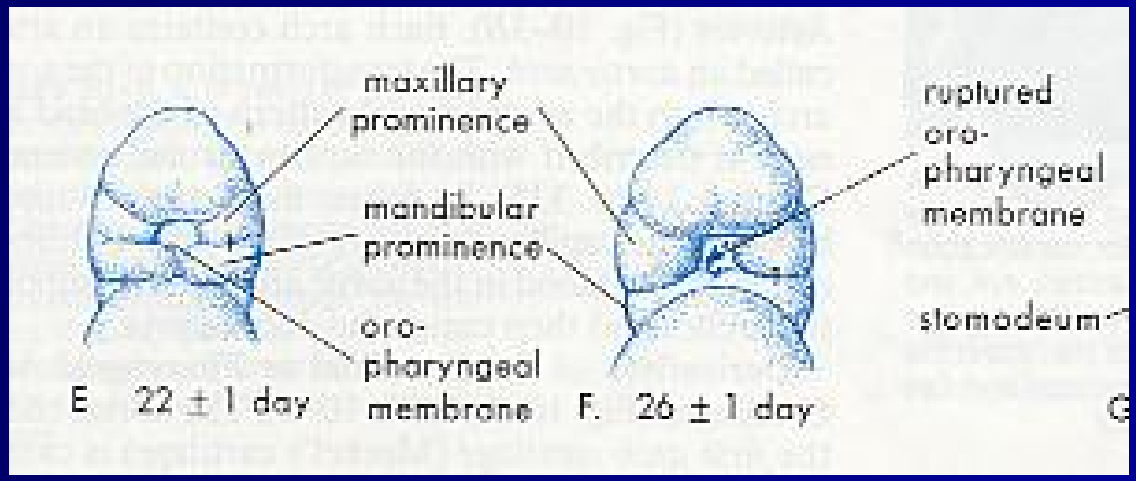
Branchial Arches

Each arch contains:

- A cartilagenous component
- A muscular component
- An aortic arch (artery)
- A nerve

First Branchial Arch

- Divides early into 2 portions
  - **Maxillary** process dorsally (maxilla, zygoma, squamous temporal bone)
  - **Mandibular** process ventrally



First Branchial Arch

- Cartilage (Meckel's Cartilage)
  - Dorsal end becomes the malleus and incus
  - Intermediate portion regresses, but the perichondrium forms:
    - Anterior ligament of the malleus
    - Sphenomandibular ligament
  - Ventral portion forms the mandible

First Branchial Arch

Muscular component

- Muscles of mastication (temporalis, masseter, med & lat pterygoids)
- Accessorymuscles of mastication (mylohyoid, ant belly of digastric)
- Tensor tympani
- Tensor veli palatini

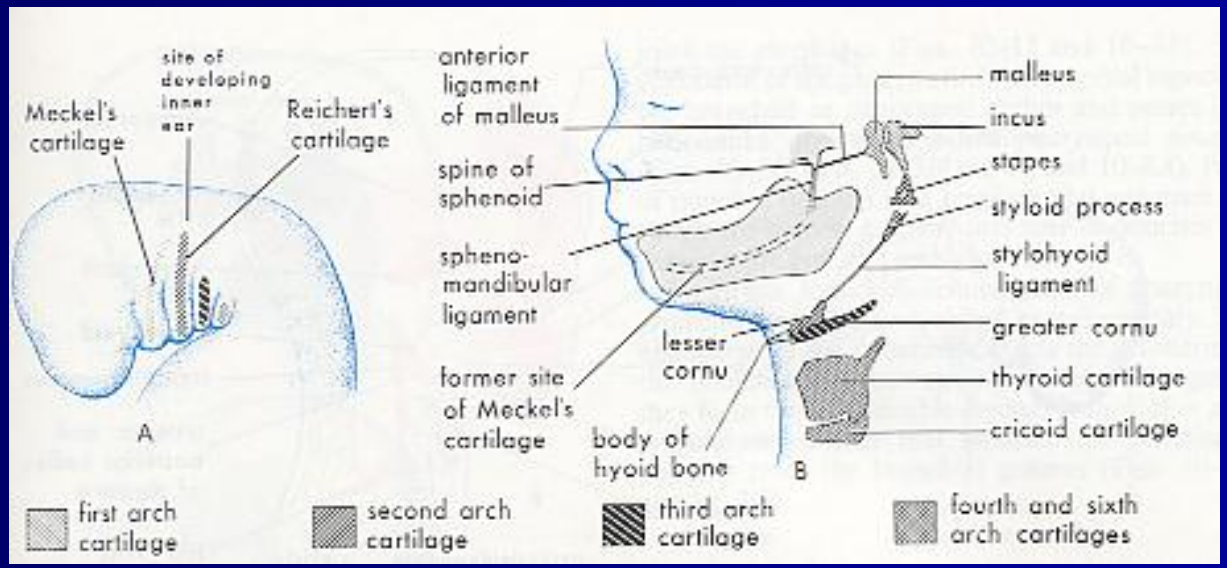
First Branchial Arch

Aortic arch

- Maxillary artery

Nerve

- Trigeminal nerve (CN V)



Second Branchial Arch

Cartilage (Reichert's Cartilage)

- Dorsal end becomes stapes (except footplate) and styloid process
- Intermediate portion regresses and perichondrium forms the stylohyoid ligament
- Ventral end forms the lesser cornu of the hyoid and the upper half of the hyoid bone

Second Branchial Arch

Muscular Component

- Migrates over superficial face to form the muscles of facial expression
- Stapedius muscle
- Stylohyoid muscle
- Posterior belly of digastric

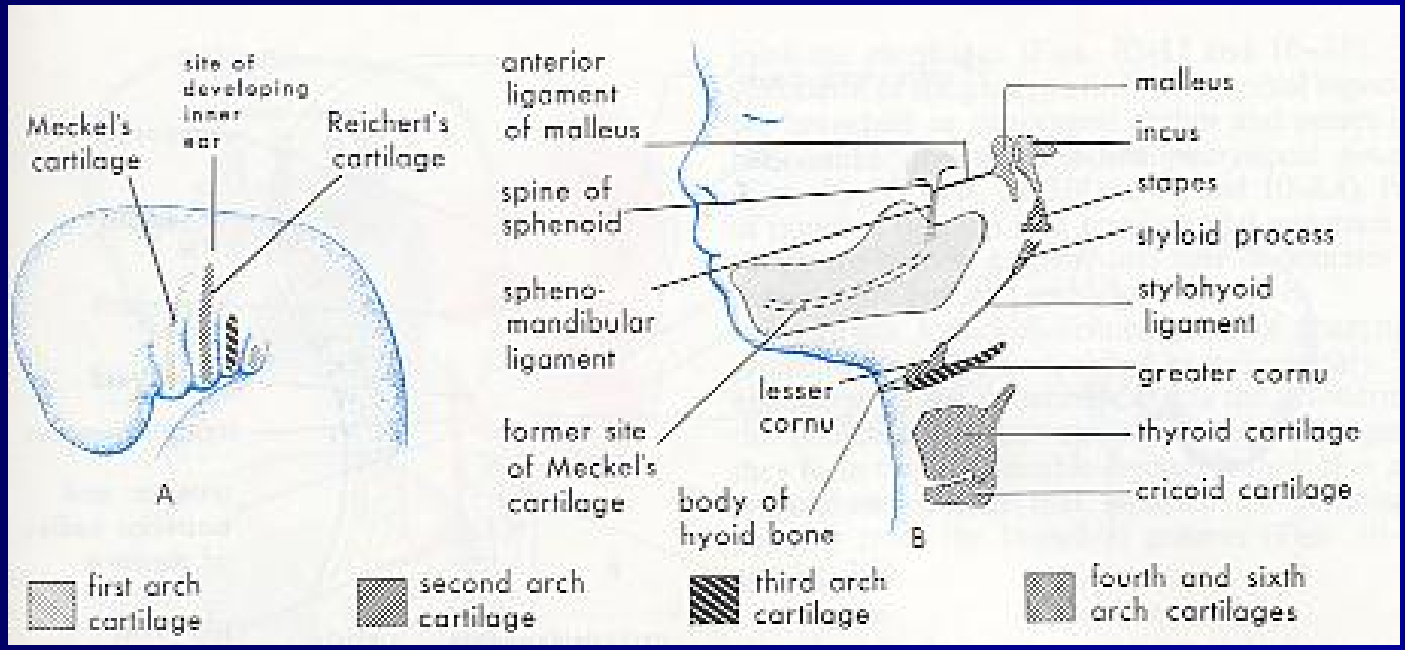
Aortic Arch

- Hyoid artery
- Stapedial artery

Second Branchial Arch

Nerve

- Facial Nerve (CN VII)



Third Branchial Arch

Cartilage

- Located ventrally and forms the lower half of the body of the hyoid and the greater cornu

Muscular Component

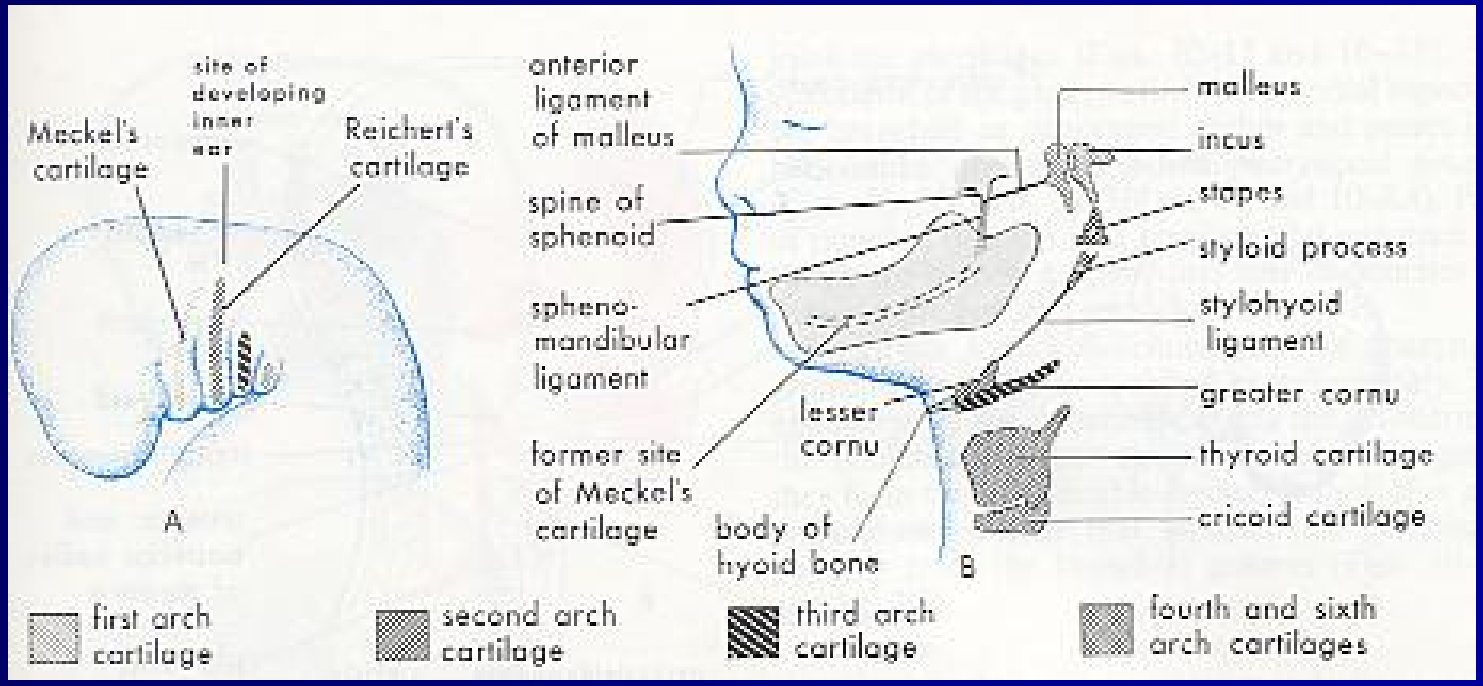
- Only one muscle: stylopharyngeus

Aortic Arch

- Common carotid, external carotid, proximal internal carotid

Third Branchial Arch

- Nerve
  - Glossopharyngeal (CN IX)



Fourth Branchial Arch

Cartilage
- Thyroid cartilage

Muscular Component
- The 3 pharyngeal constrictors
- Cricothyroid muscle

Aortic Arch
- Left: Aortic arch
- Right: Right subclavian

Nerve
- Superior laryngeal branch of vagus (CN X)

Sixth Branchial Arch

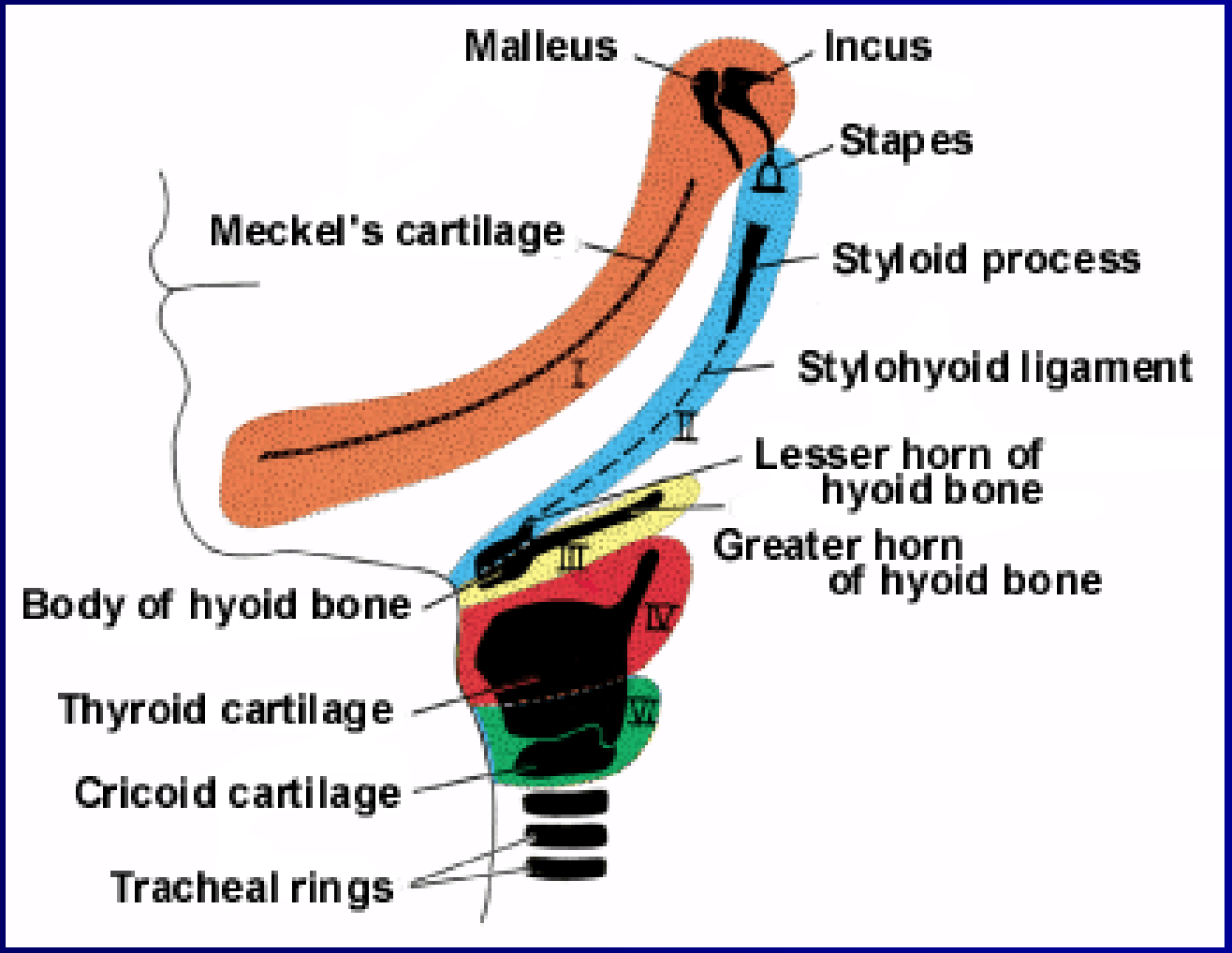
Cartilage
- Cricoid cartilage, arytenoid, corniculate and cuneiform

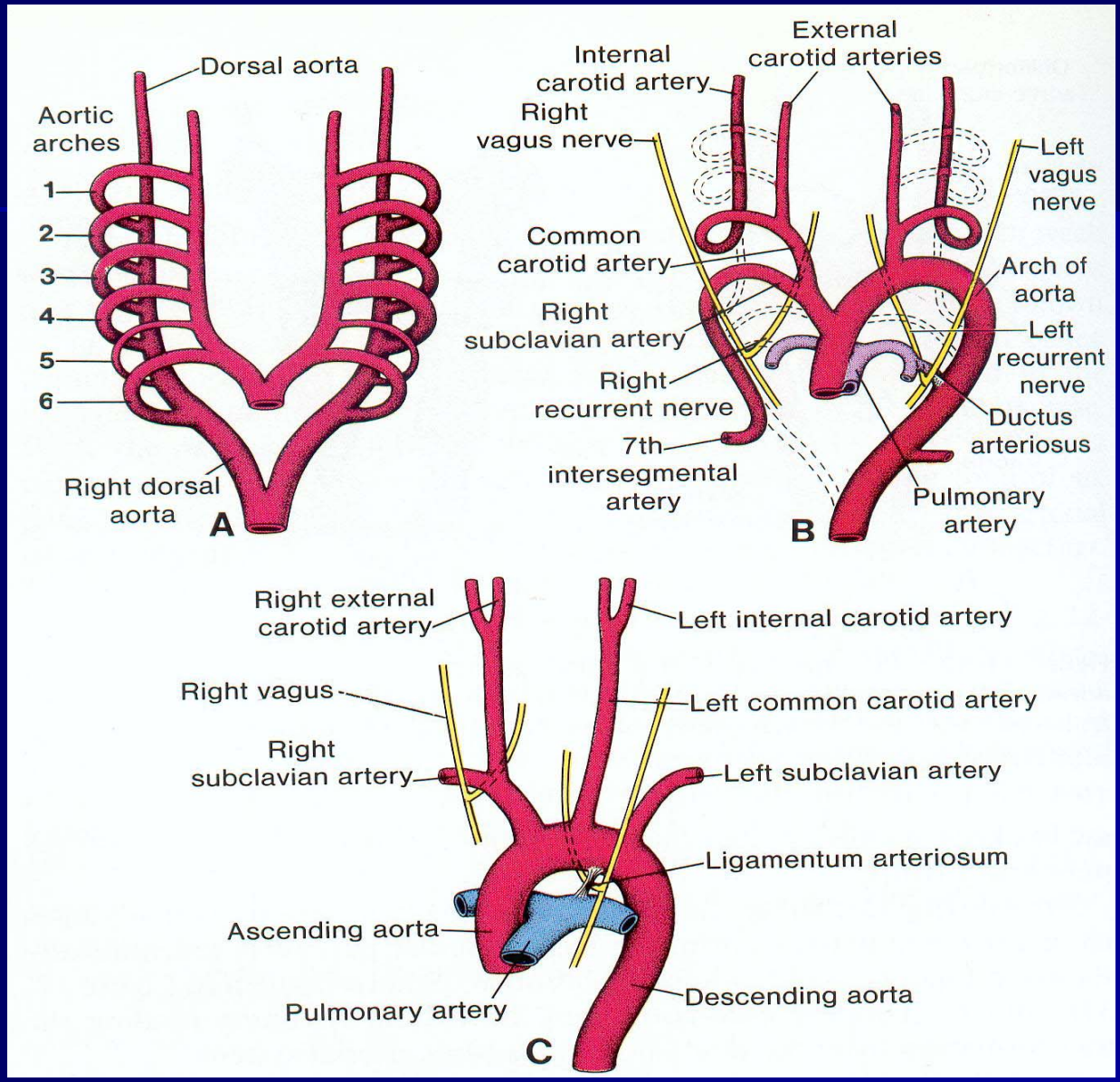
Muscular Component
- Intrinsic muscles of larynx

Aortic Arch
- Left: pulmonary artery, ductus arteriosus
- Right: pulmonary artery, distal end degenerates

Nerve
- Recurrent laryngeal nerve of vagus (CN X)

Branchial Arches Summary

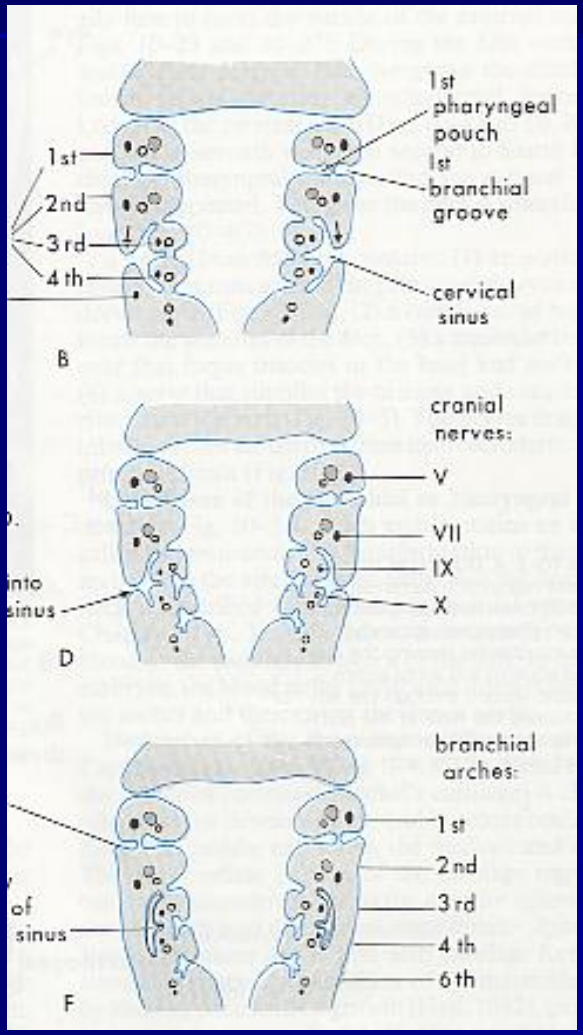



Dorsal aorta
Aortic arches
1
2
3
4
5
6
Right dorsal aorta
A
Internal carotid artery
External carotid arteries
Right vagus nerve
Left vagus nerve
Common carotid artery
Arch of aorta
Right subclavian artery
Left recurrent nerve
Right recurrent nerve
Ductus arteriosus
7th intersegmental artery
Pulmonary artery
B
Right external carotid artery
Left internal carotid artery
Right vagus
Left common carotid artery
Right subclavian artery
Left subclavian artery
Ligamentum arteriosum
Ascending aorta
Pulmonary artery
Descending aorta
C

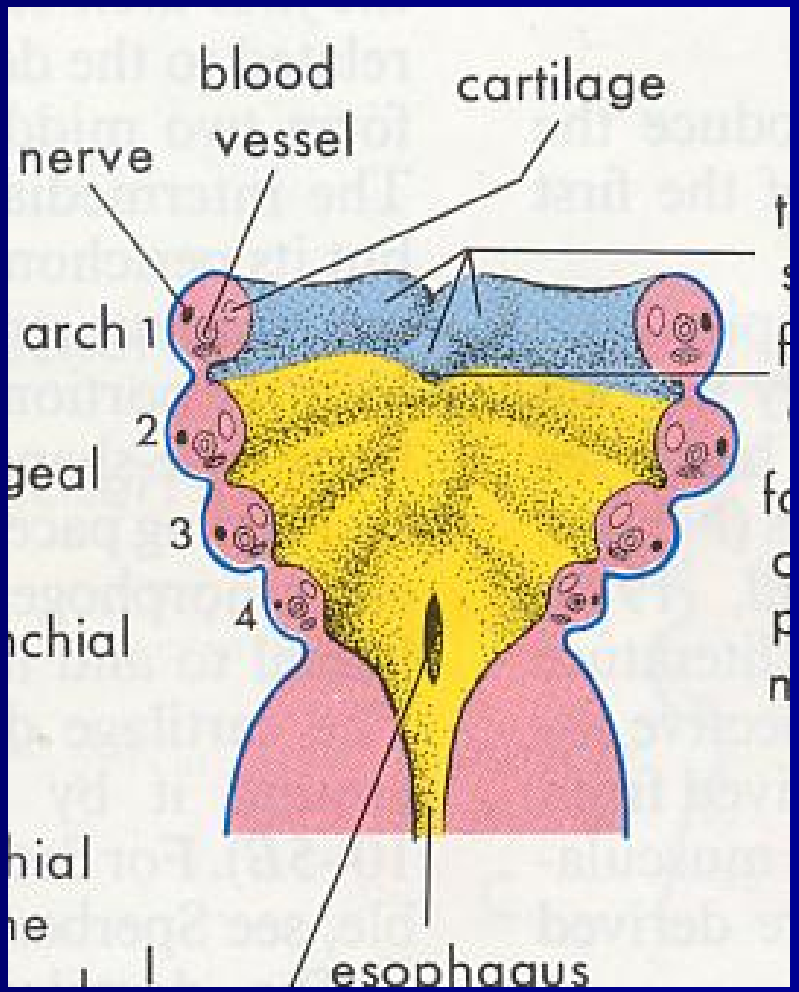
Branchial Clefts (or grooves)

- 4 clefts
- The 2nd to 4th clefts become buried by the overgrowth of the 2nd arch to form the cervical sinus
- Cervical sinus has dissapeared by week 7
- The first cleft persists and invades the mesenchyme opposite the first pouch
- This becomes the EAC and ectoderm of the TM

Branchial Clefts



Pharyngeal Pouches



1st Pharyngeal Pouch

- Elongates into the tubotympanic recess
- The distal part contacts the 1st pharyngeal cleft and forms the inner lining of the TM
- The tubotympanic recess becomes the tympanic cavity and mastoid antrum
- Connection of the recess with the pharynx becomes the eustachian tube

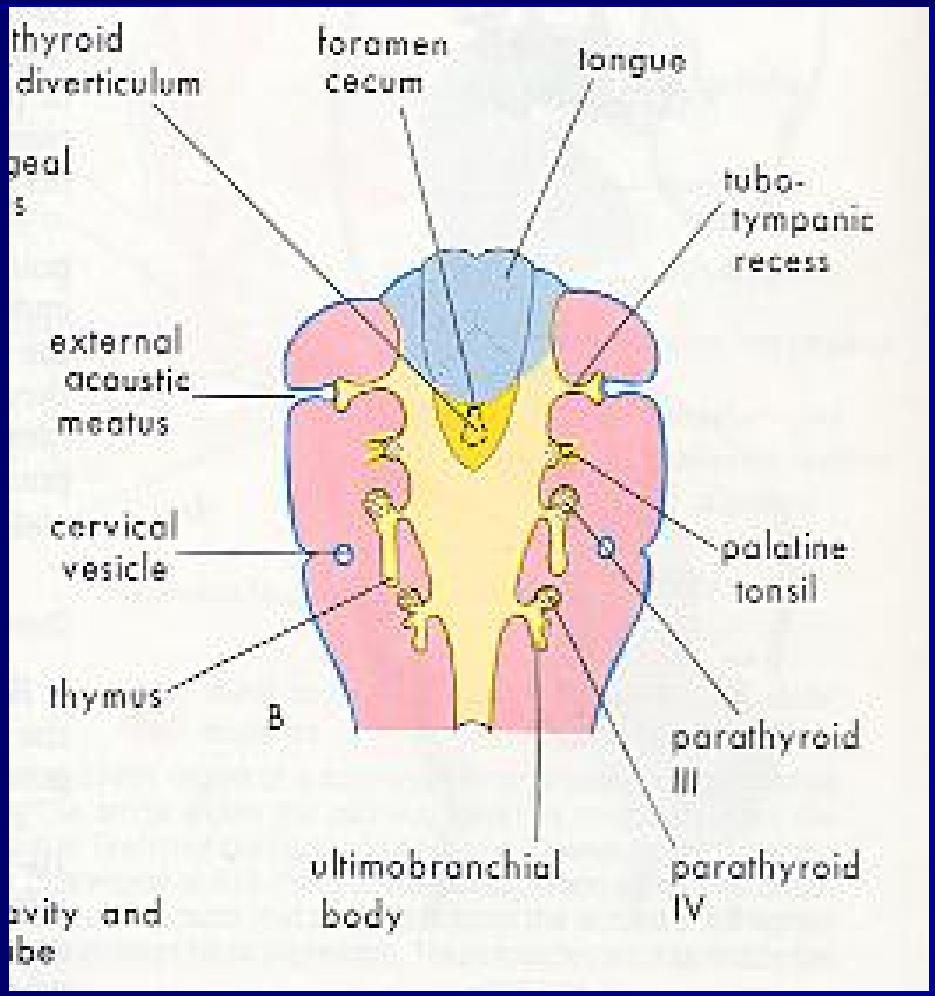
2nd Pharyngeal Pouch

- Forms the tonsillar fossa
- Endoderm forms the surface epithelium and lining of tonsillar crypts
- At 20 weeks lymphoid tissue invades the endoderm and forms the palatine tonsils

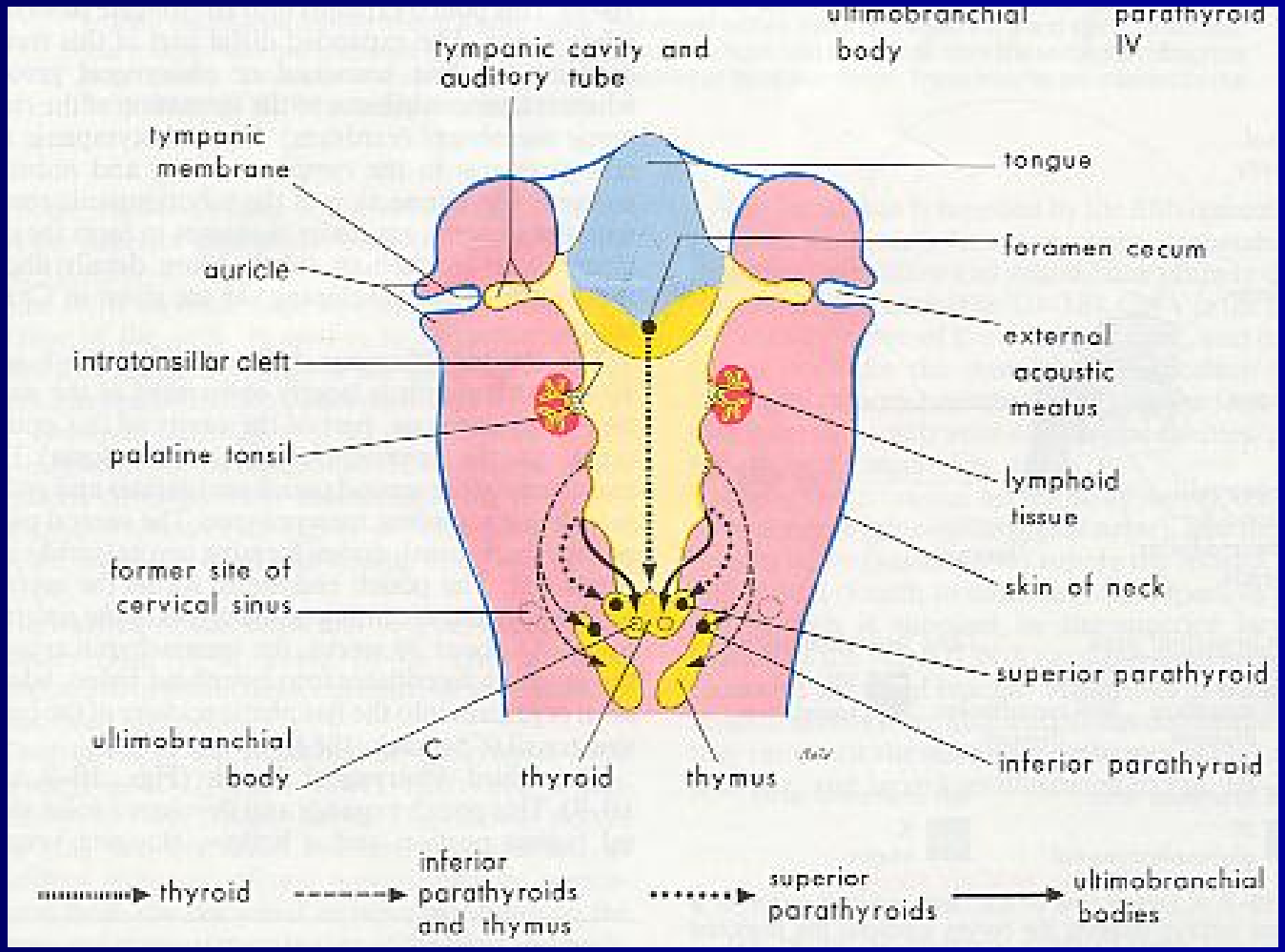
3rd Pharyngeal Pouch

- Forms 2 diverticula: dorsal and ventral
- Endoderm of **dorsal** diverticula: inf parathyroid
- Endoderm of **ventra**l diverticula: lobule of thymus
- These diverticula become detached from the wall and migrate caudally.
- Thymus comes to lie in the superior mediastinum

Pharyngeal Pouches



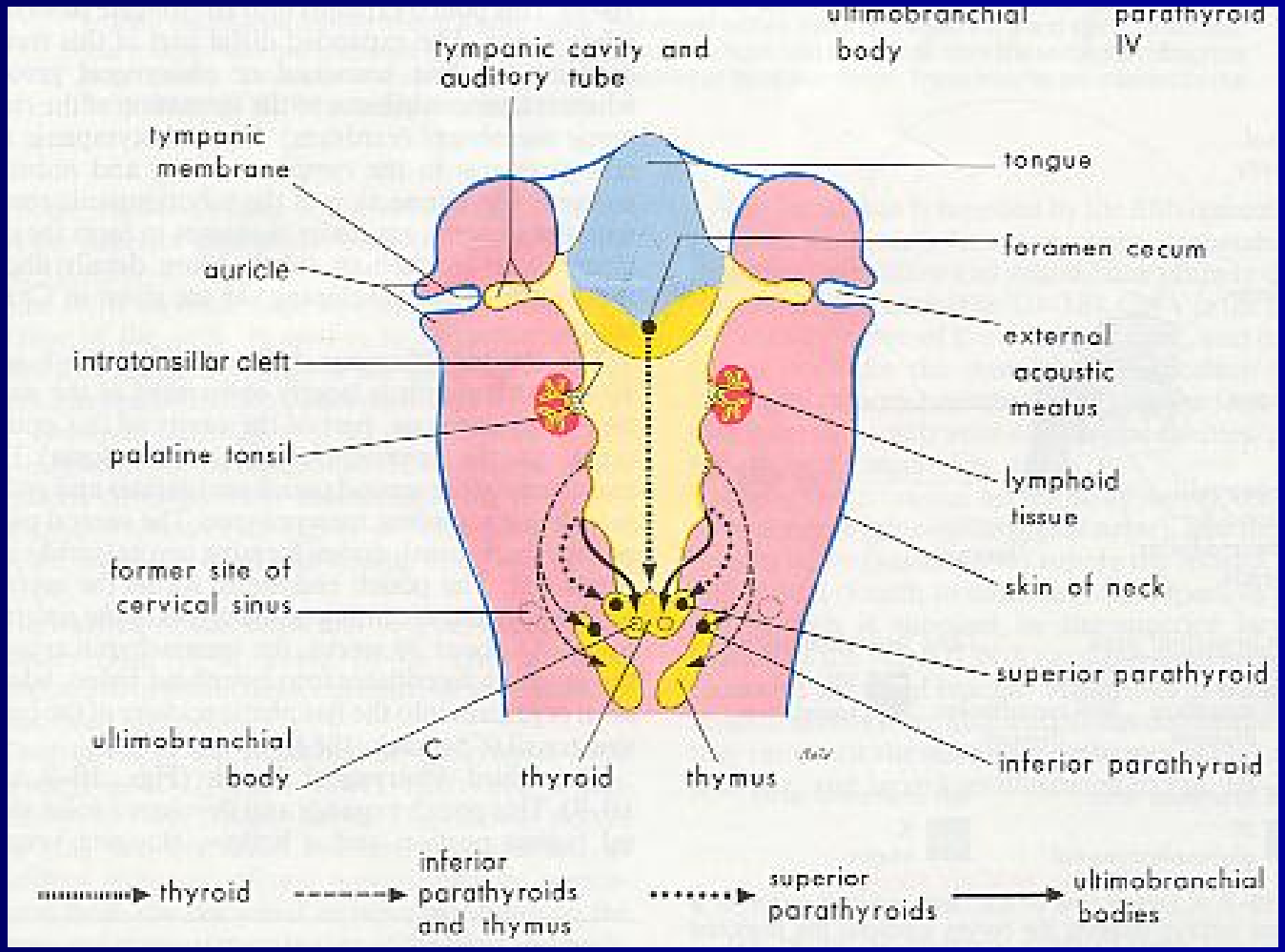
Pharyngeal Pouches



4th Pharyngeal Pouch

- Also develops dorsal and ventral diverticula
- The **dorsal** bud becomes the sup. parathyroid
- The **ventral** bud becomes the **Ultimobranchial body**
- The ultimobranchial body fuses with the thyroid gland and disseminates within it to give the parafollicular C cells which produce calcitonin

Pharyngeal Pouches



Pharyngeal Pouches

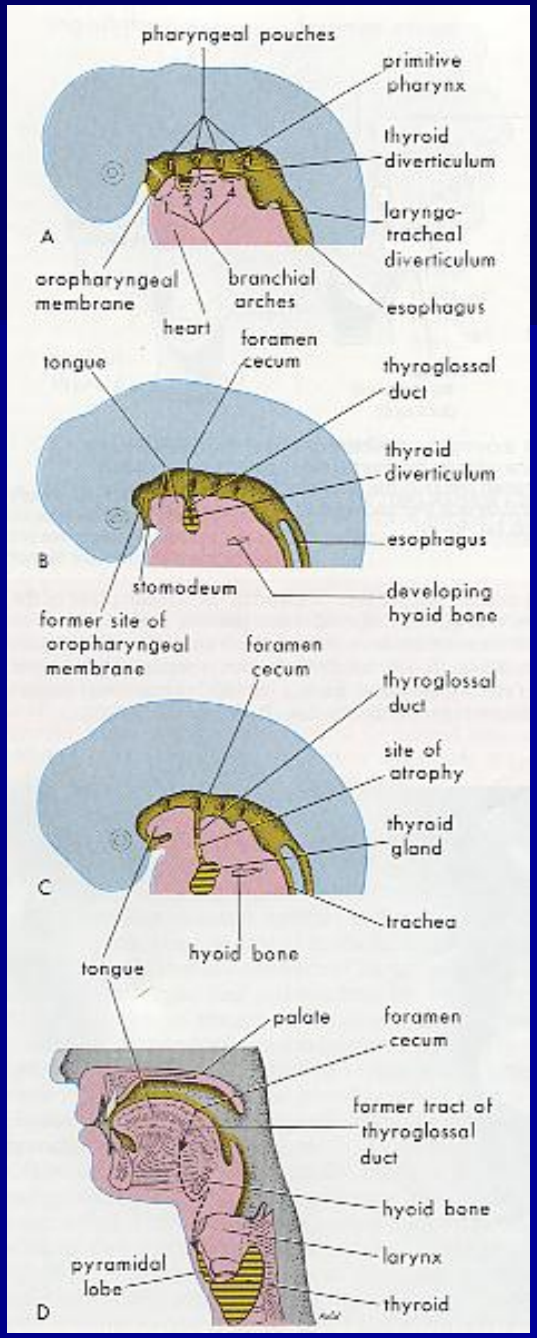
- 5th pouch never develops
- Controversy re: originof ultimobranchial body

Thyroid

- In 4th week begins as endodermal thickening in floor of primitive pharynx
- The thickening becomes an outpouching: thyroid diverticulum
- Thyroid descends anterior to hyoid and thyroid cartilage
- Connected to tongue by thyroglossal duct
- Week 7: Thyroid reaches final position
  Thyroglossal duct has degenerated

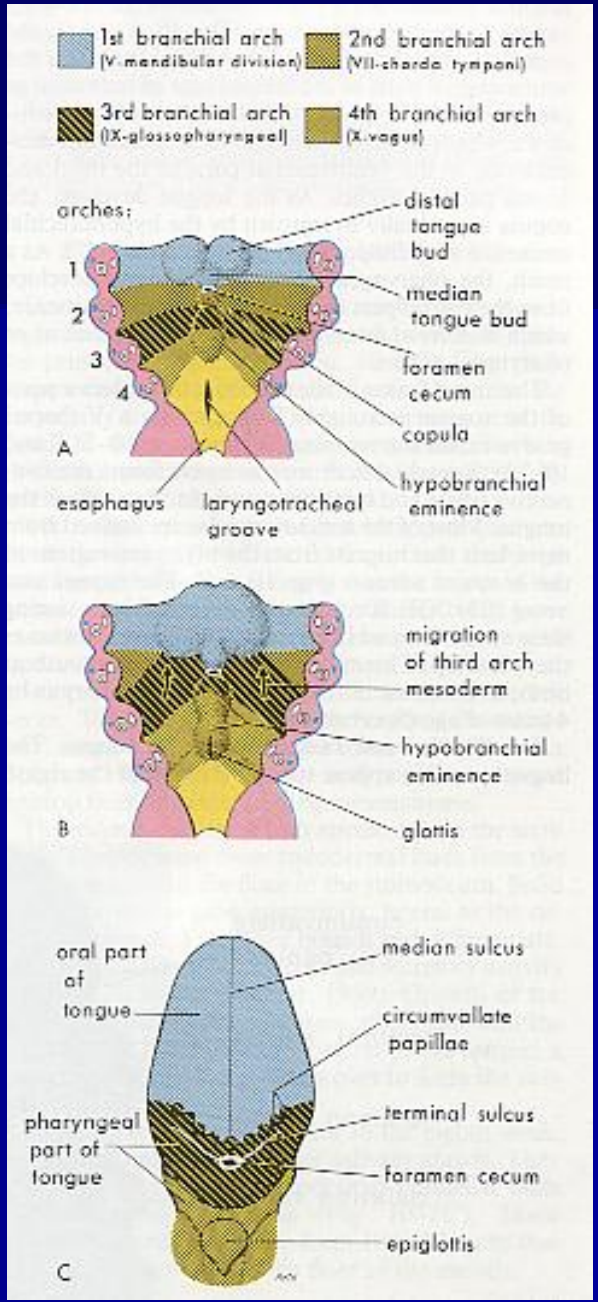
- Pyramidal lobe: Persistence of distal end of thyroglossal duct
  Present in 50% of people

Thyroid


pharyngeal pouches
primitive pharynx
thyroid diverticulum
laryngo-tracheal diverticulum
esophagus
A
oropharyngeal membrane
heart
branchial arches
tongue
foramen cecum
thyroglossal duct
thyroid diverticulum
esophagus
B
stomodeum
developing hyoid bone
former site of oropharyngeal membrane
foramen cecum
thyroglossal duct
site of atrophy
thyroid gland
C
trachea
hyoid bone
tongue
palate
foramen cecum
former tract of thyroglossal duct
hyoid bone
larynx
pyramidal lobe
thyroid
D

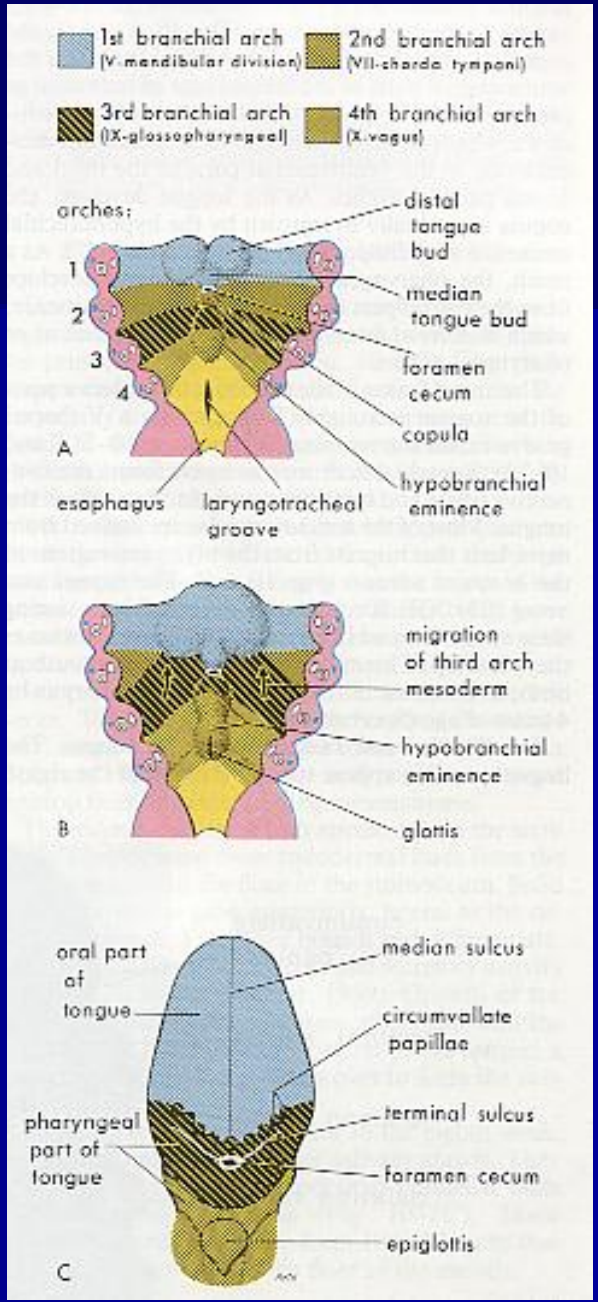
Tongue

- 4th week: elevation on floor of pharynx, just rostral to foramen cecum: **Median Tongue Bud** (Tuberculum impar)
- **Distal Tongue Buds** develop just lateral to median tongue bud
- Both of the above originate in mesenchyme of first branchial arch



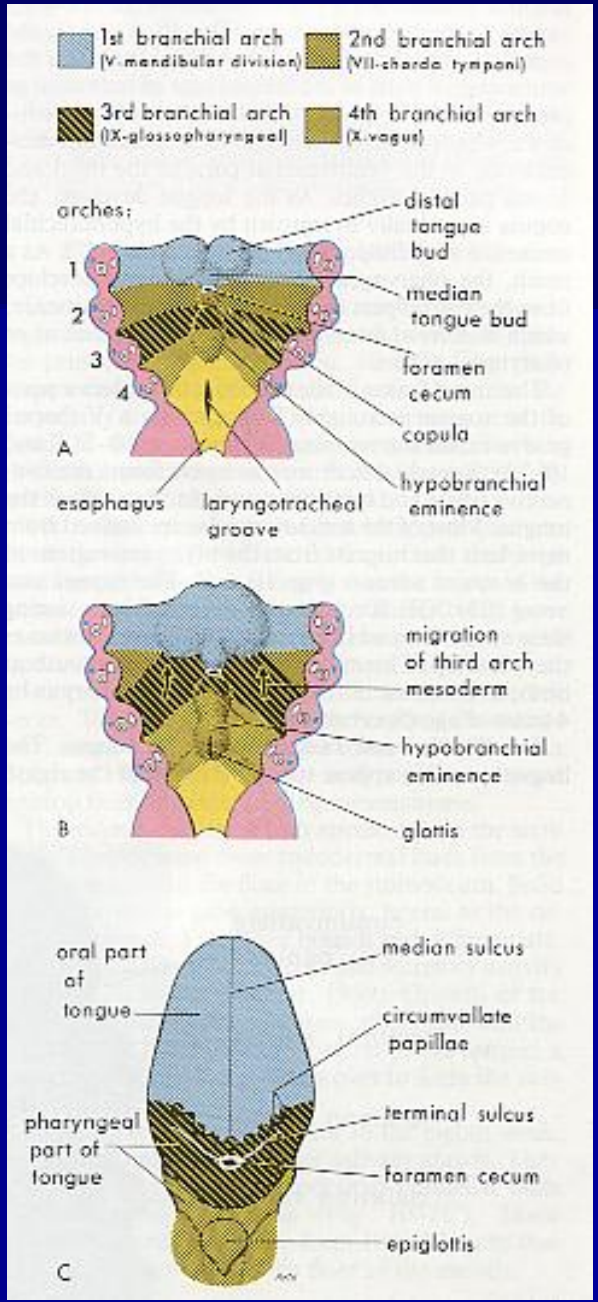
Tongue

- Distal tongue buds overgrow the median tongue bud and merge with each other
- These form the ant 2/3 of the tongue
- Median tongue bud forms no adult structure



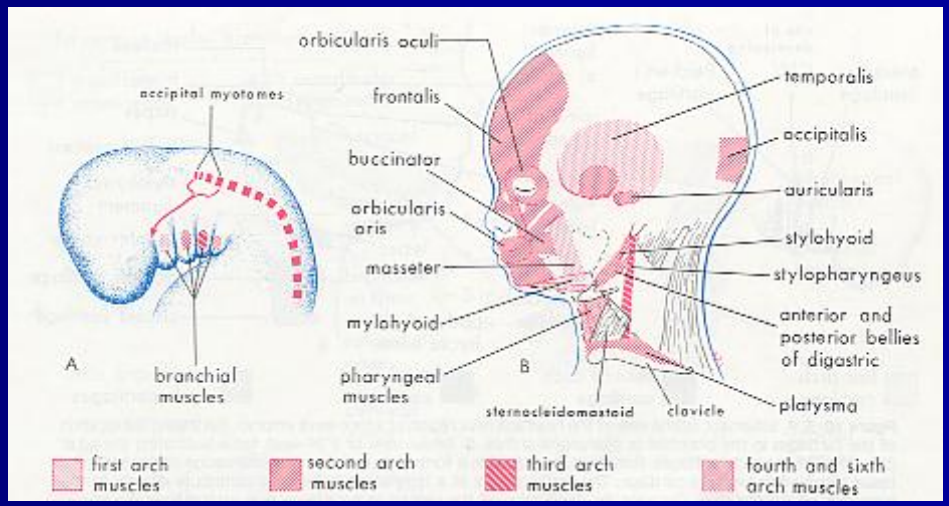
Tongue

- At same time 2 elevations develop caudal to foramen cecum:
    1. Copula: from 2nd arch
    2. Hypobranchial emminence: from 3rd & 4th arches
- The hypobranchial emminence overgrows the copula which disappears
- The post 1/3 of the tongue is formed by the rostral part of the hypobranchial emminence (Arch 3)
- Caudal part of hypobranchial emminence (Arch 4) forms the epiglottis



Tongue

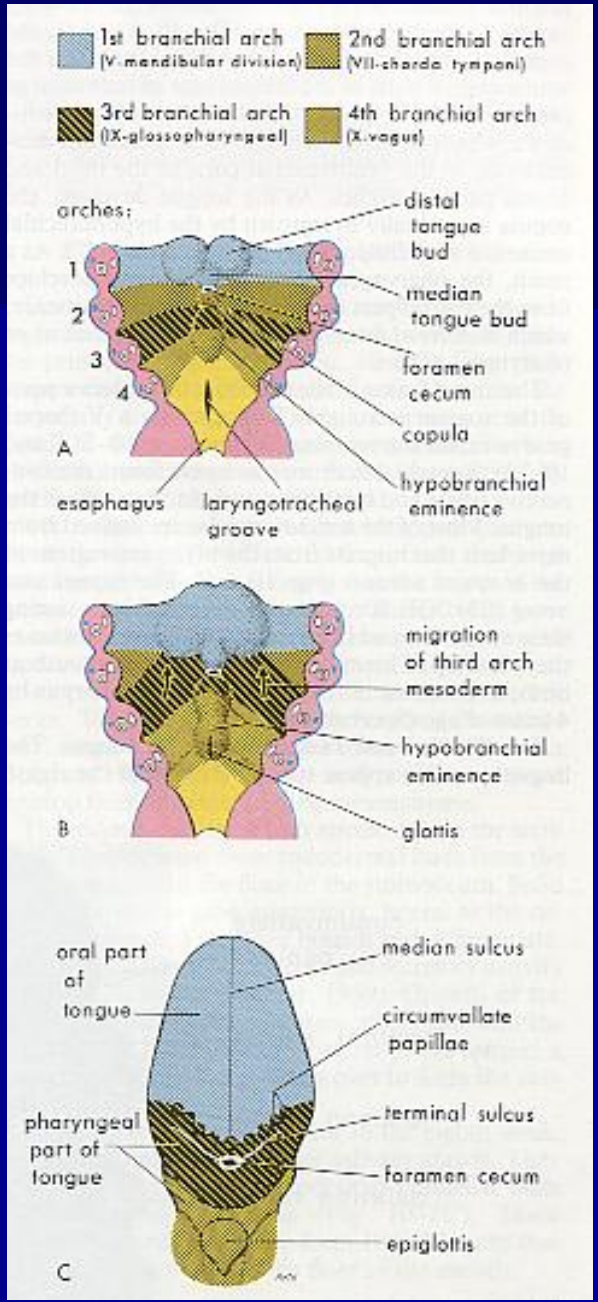
- Branchial mesenchyme forms the soft tissue, vascular and lymphatics of the tongue.
- Tongue muscles originate from the occipital somites which bring with them innervation (CN XII)



Tongue

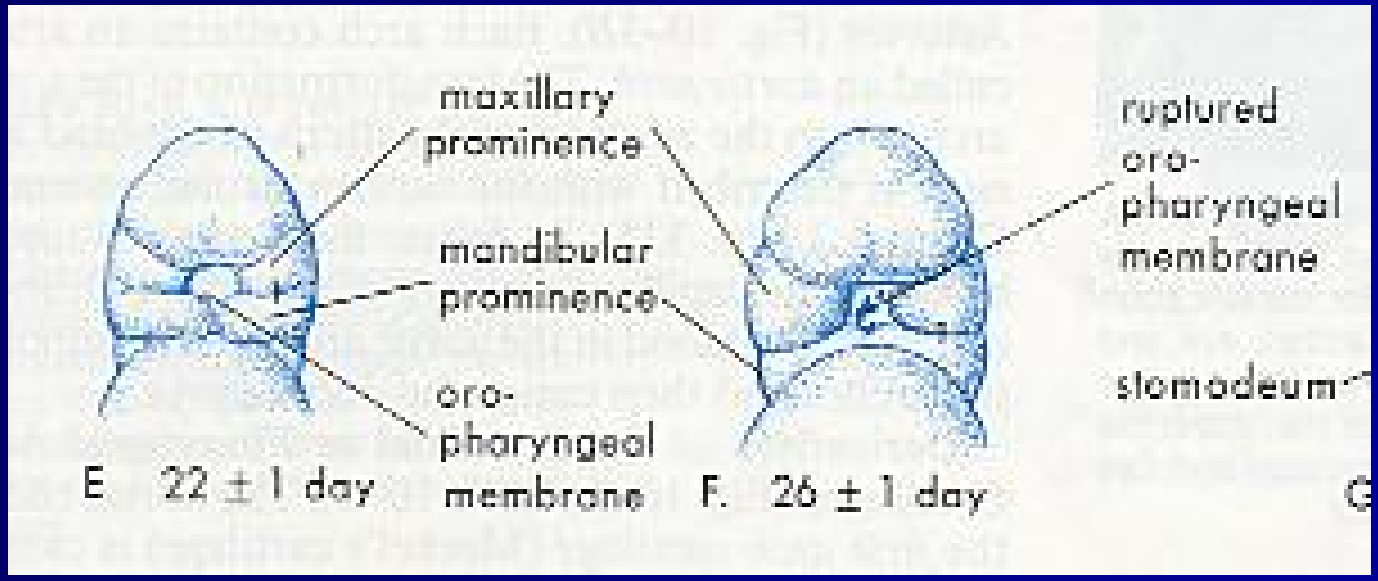
Innervation to tongue:

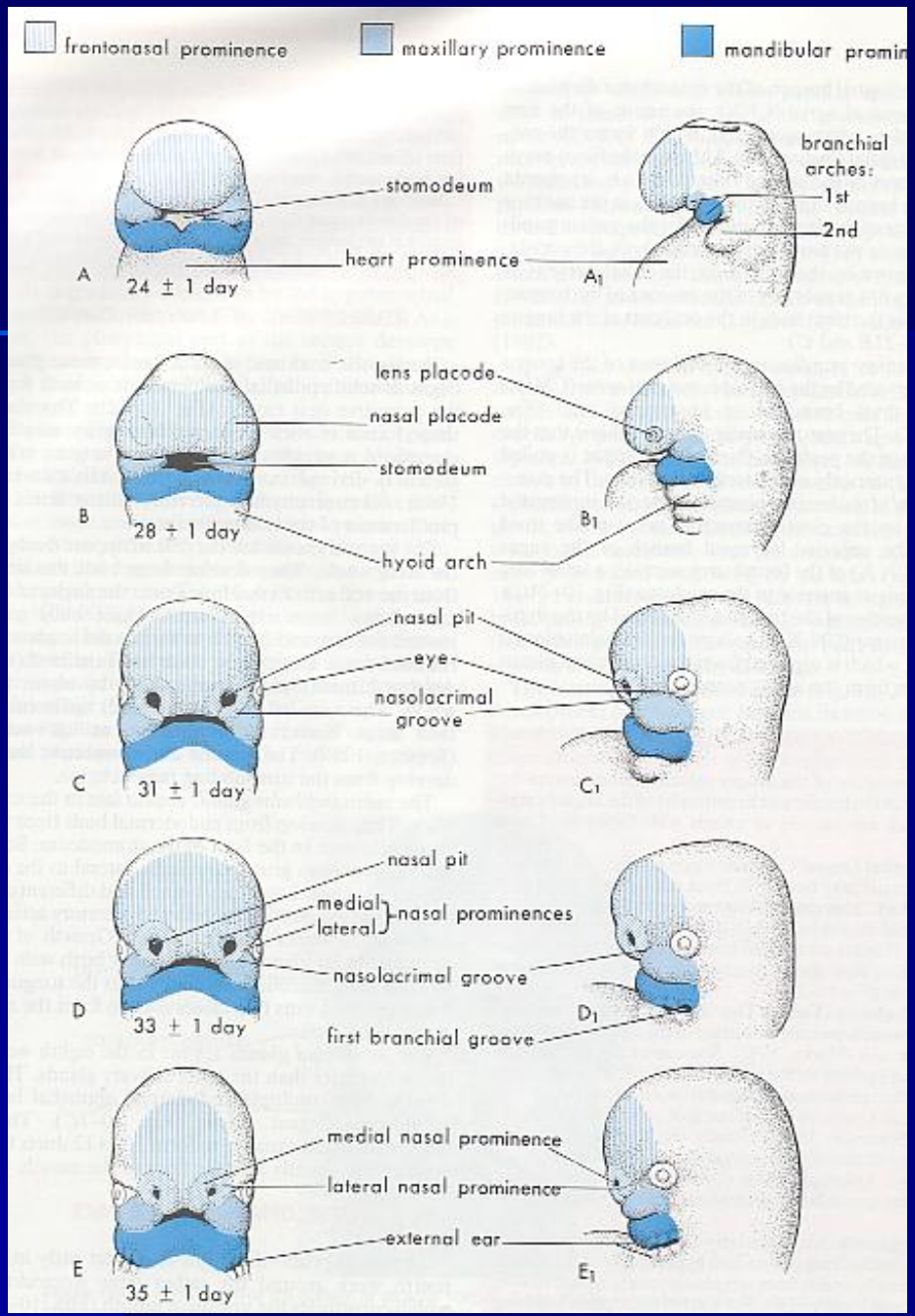
- Ant 2/3: CN V
- Post 1/3: CN IX



Development of the Face

- Five facial primordia contribute to development of the face:
  - The frontonasal prominence
  - Paired Maxillary prominences
  - Paired Mandibular prominences





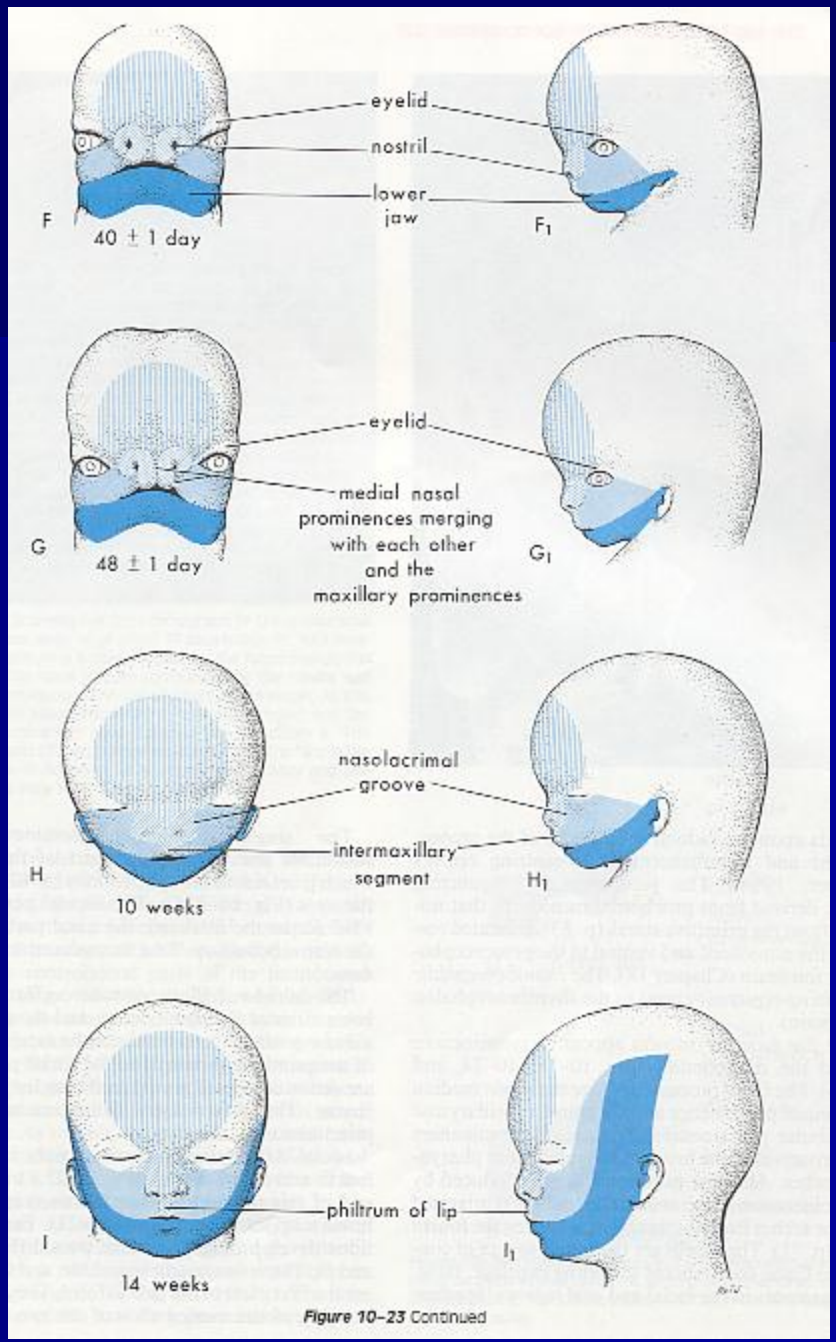
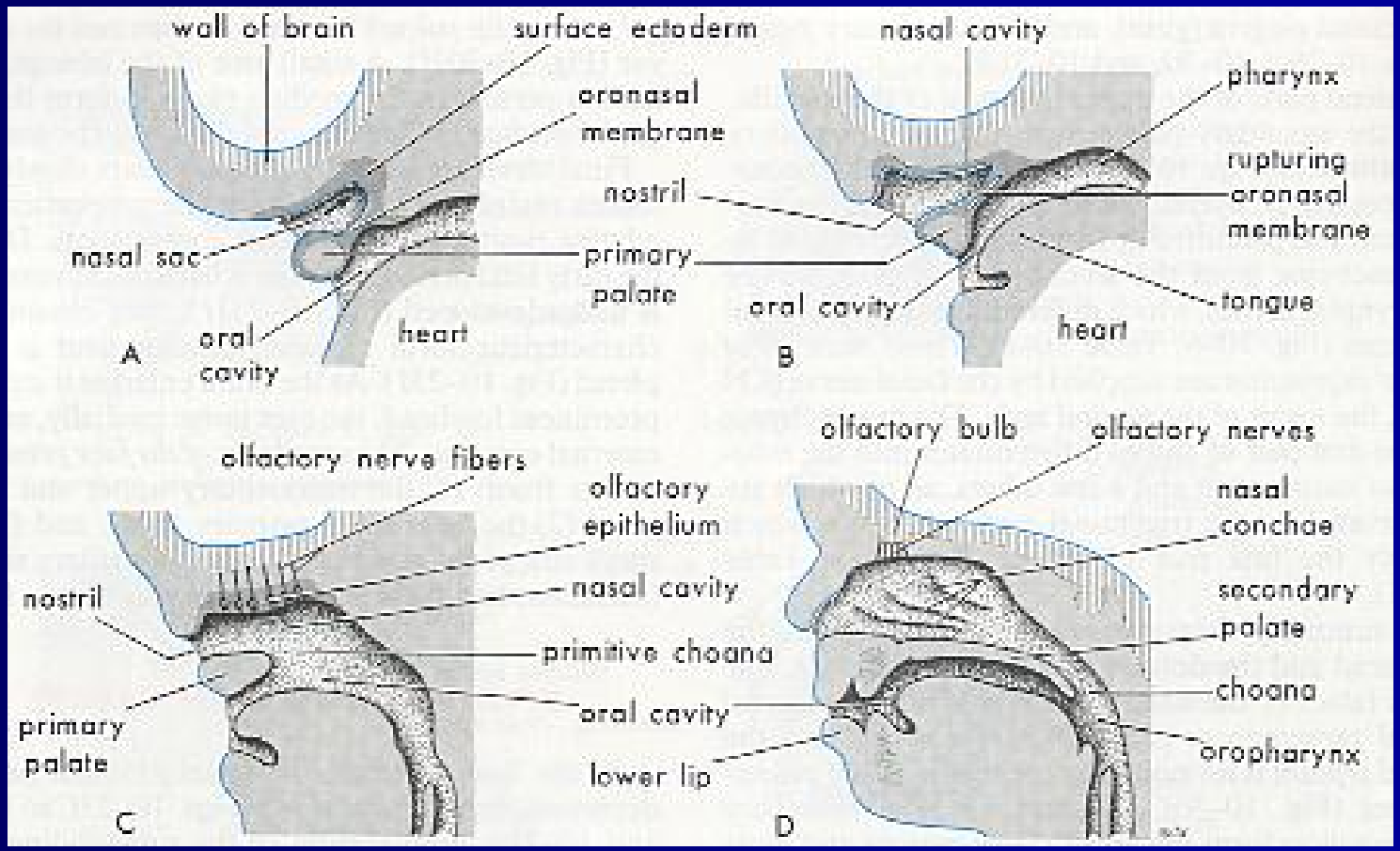


Figure 10–23 Continued

Development of the Face

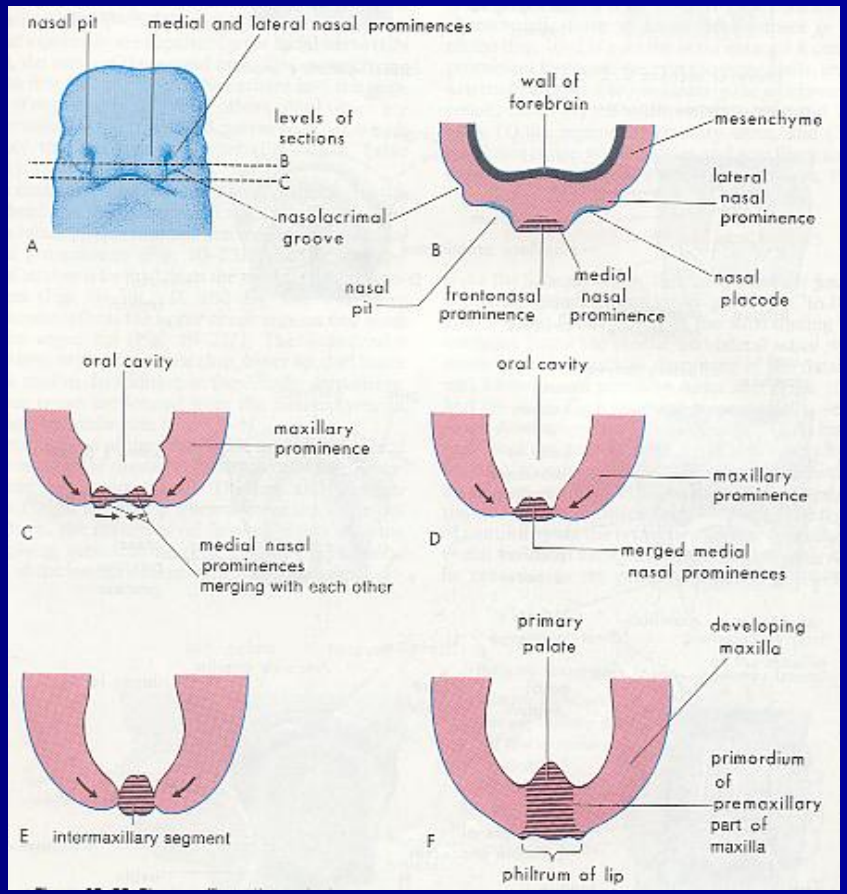
- 4th week: thickening of ectoderm in the ventrolateral parts of the FNP: **Nasal Placodes**
- Mesenchyme on the edges of the placodes proliferates to form: **medial and lateral nasal prominences**
- As a result the nasal placodes now lie in a depression called **nasal pits** which enlarge dorsally to form the nasal cavities.
- These nasal cavities are separated from the oral cavity by the **oronasal membranes** which rupture to form the primitive choana

Development of the Face



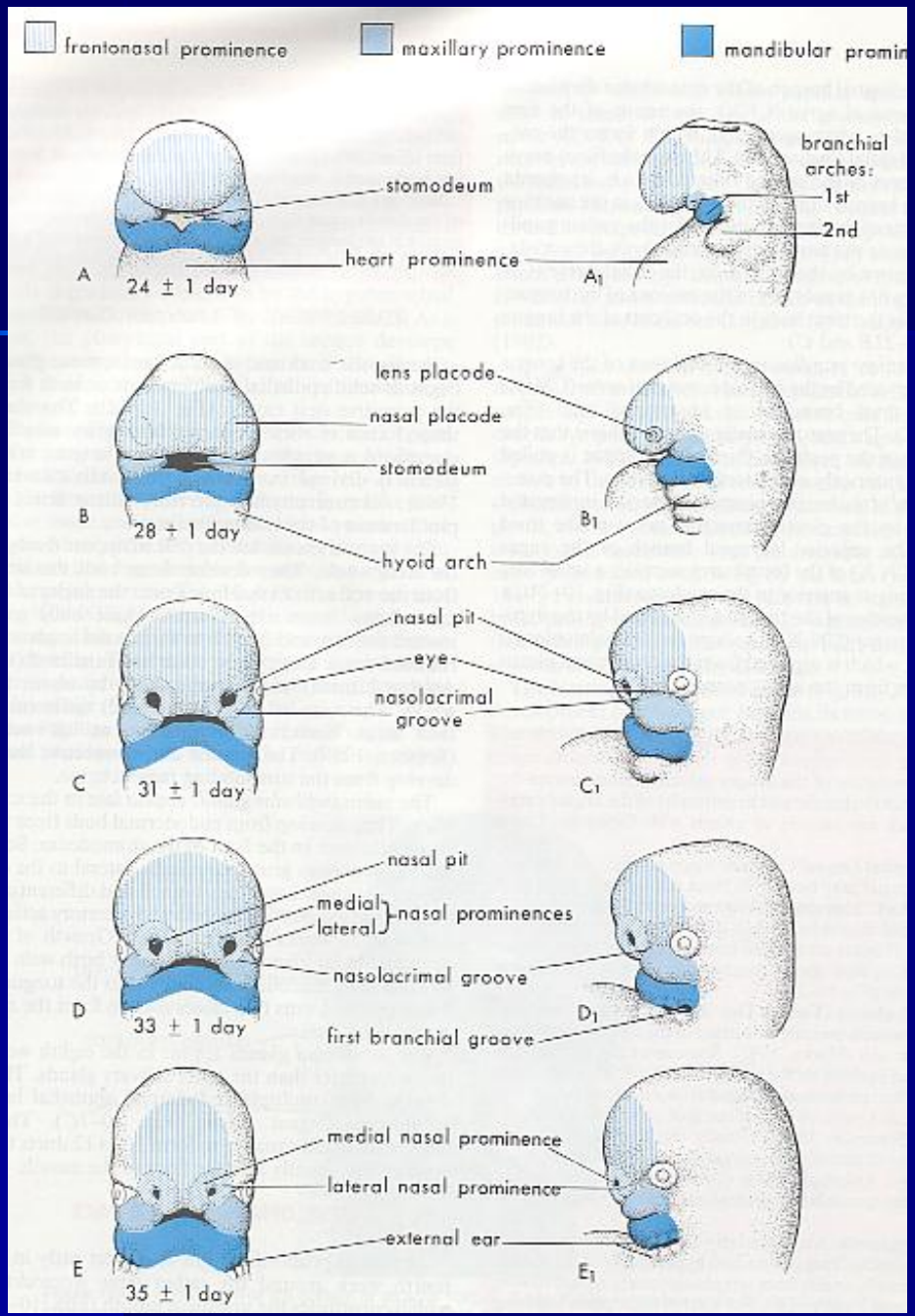
Development of the Face

- Growth of maxillary prominences pushes medial nasal prominences medially
- These fuse to form the nasal bulb, the philtrum, the premaxillary segment of the maxilla and the primitive palate



Development of the Face

- The lateral nasal prominences, which become the ala are separated from the maxillary prominences by the n**asolacrimal grooves** which become the nasolacrimal ducts.
- The sinuses form as outpouchings of the ectoderm of lateral nasal walls
- The olfactory epithelium develops from ectoderm



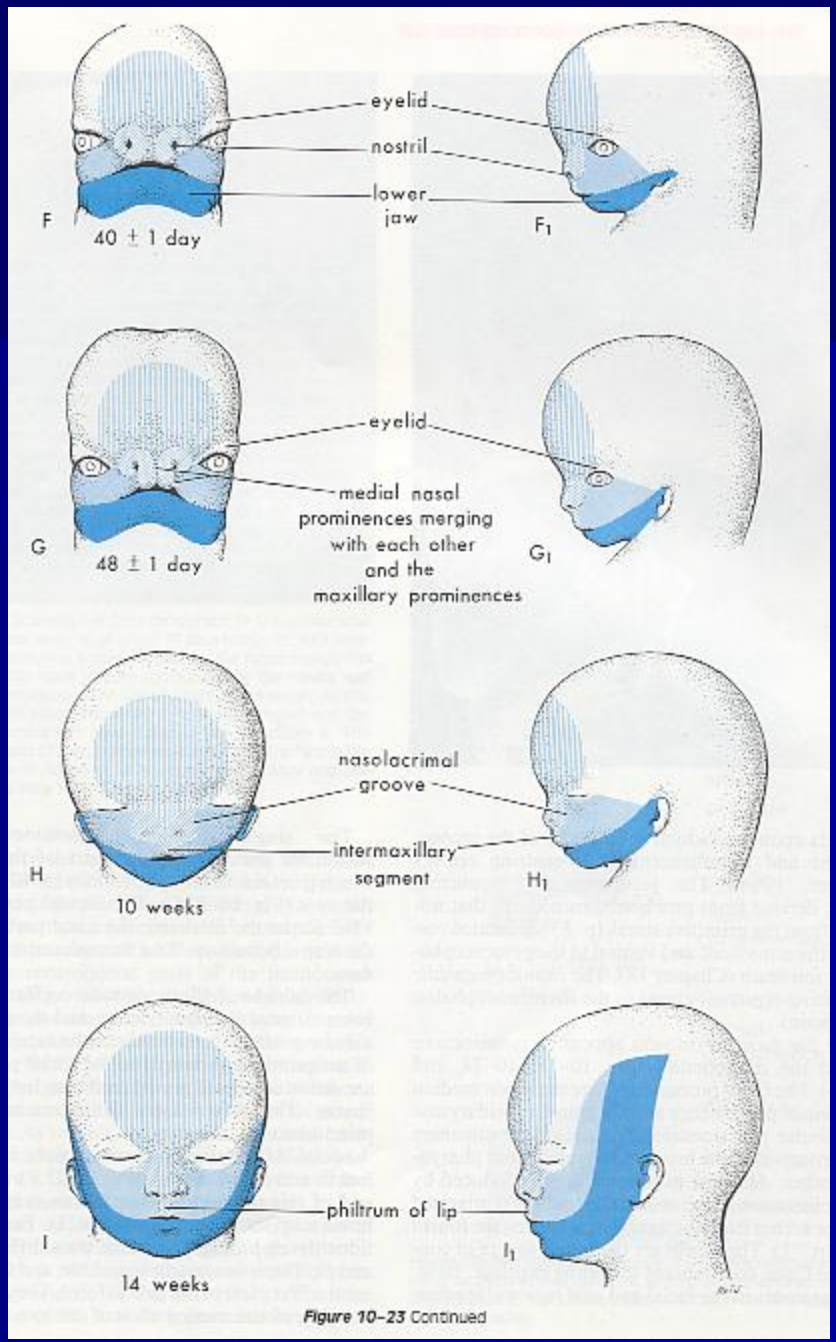


Figure 10–23 Continued

Palate

Primary and Secondary Palate

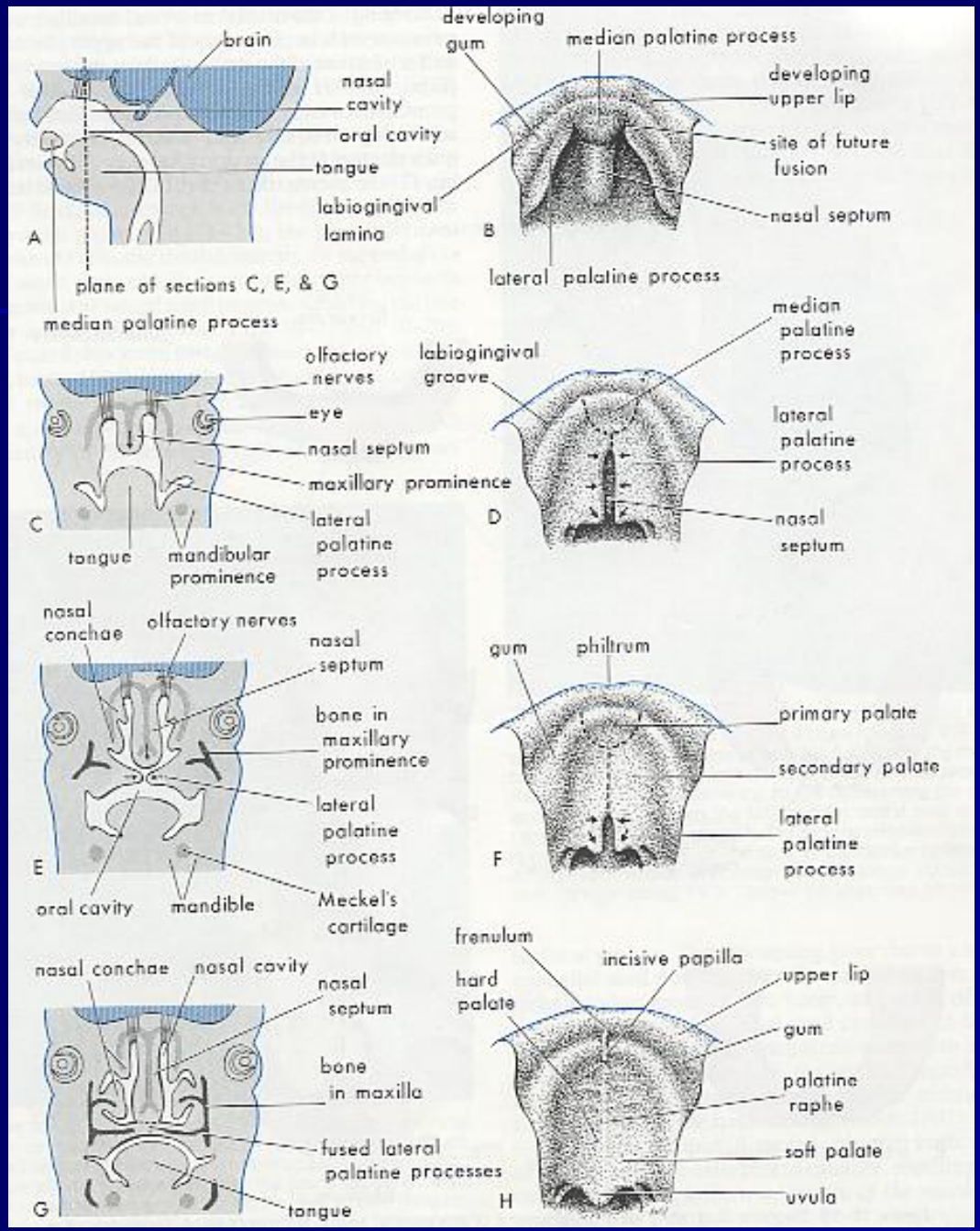
Primary Palate

- Develops from the fusion of the medial nasal prominences between the maxillary prominences
- Forms the adult portion of the palate which is anterior to the incisive foramen

Palate

Secondary Palate

- Origin of the hard and soft palate
- Develops from internal projections of the maxillary prominences called the **lateral palatine processes**
- As mandible develops, the tongue drops and the palatine processes grow medially and fuse in the midline.
- They also fuse with the nasal septum and the primary palate.
- Ossification occurs in an antero-posterior direction


brain
nasal cavity
oral cavity
tongue
labiogingival lamina
A
plane of sections C, E, & G
median palatine process
developing gum
median palatine process
developing upper lip
site of future fusion
nasal septum
B
lateral palatine process
olfactory nerves
eye
nasal septum
maxillary prominence
lateral palatine process
C
tongue
mandibular prominence
labiogingival groove
median palatine process
lateral palatine process
nasal septum
D
nasal conchae
olfactory nerves
nasal septum
bone in maxillary prominence
lateral palatine process
E
oral cavity
mandible
Meckel's cartilage
gum
philtrum
primary palate
secondary palate
lateral palatine process
F
nasal conchae
nasal cavity
nasal septum
bone in maxilla
fused lateral palatine processes
G
tongue
frenulum
incisive papilla
hard palate
upper lip
gum
palatine raphe
soft palate
uvula
H

The Ear

Inner Ear

- 4th week: thickening in surface ectoderm called **otic placode**
- Invaginates into underlying mesenchyme and detaches from ectoderm: now called **otic vesicle**
- Otic vesicle divides into 2 regions: utricular portion and saccular portion

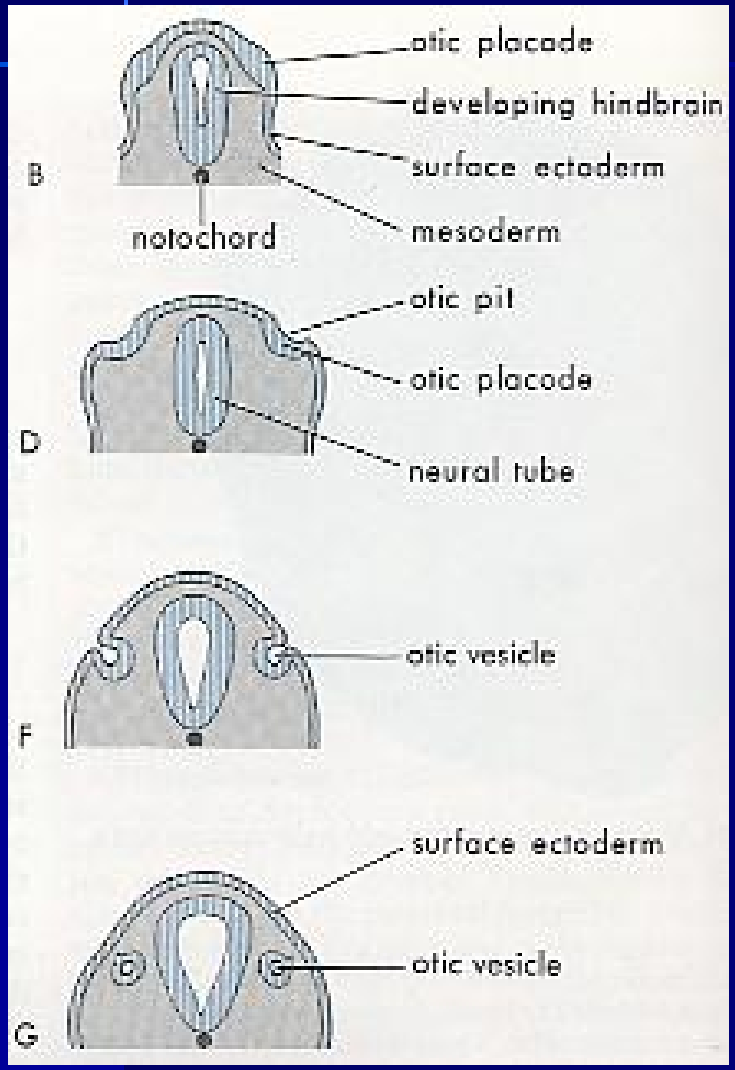
Utricular portion

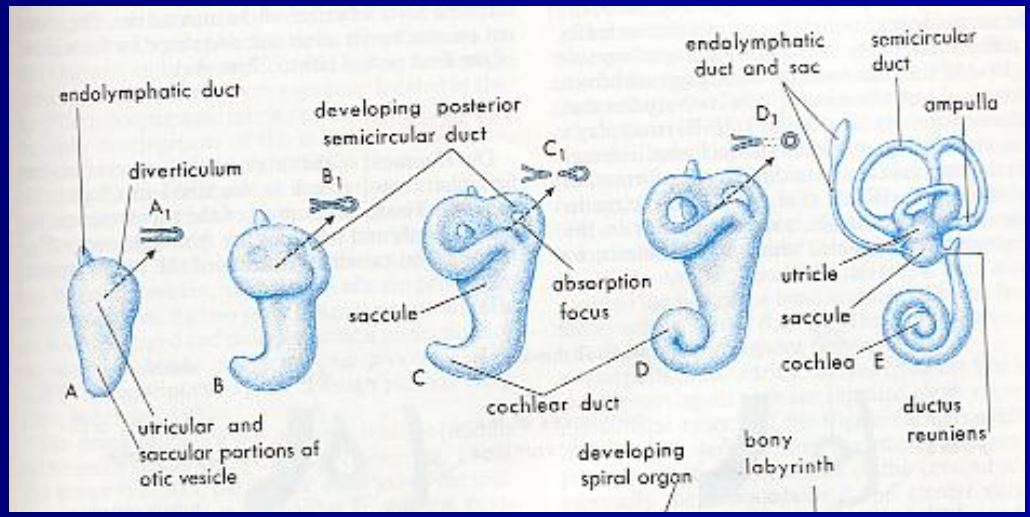
- Utricle, semicircular canals and endolymphatic duct

Saccular portion

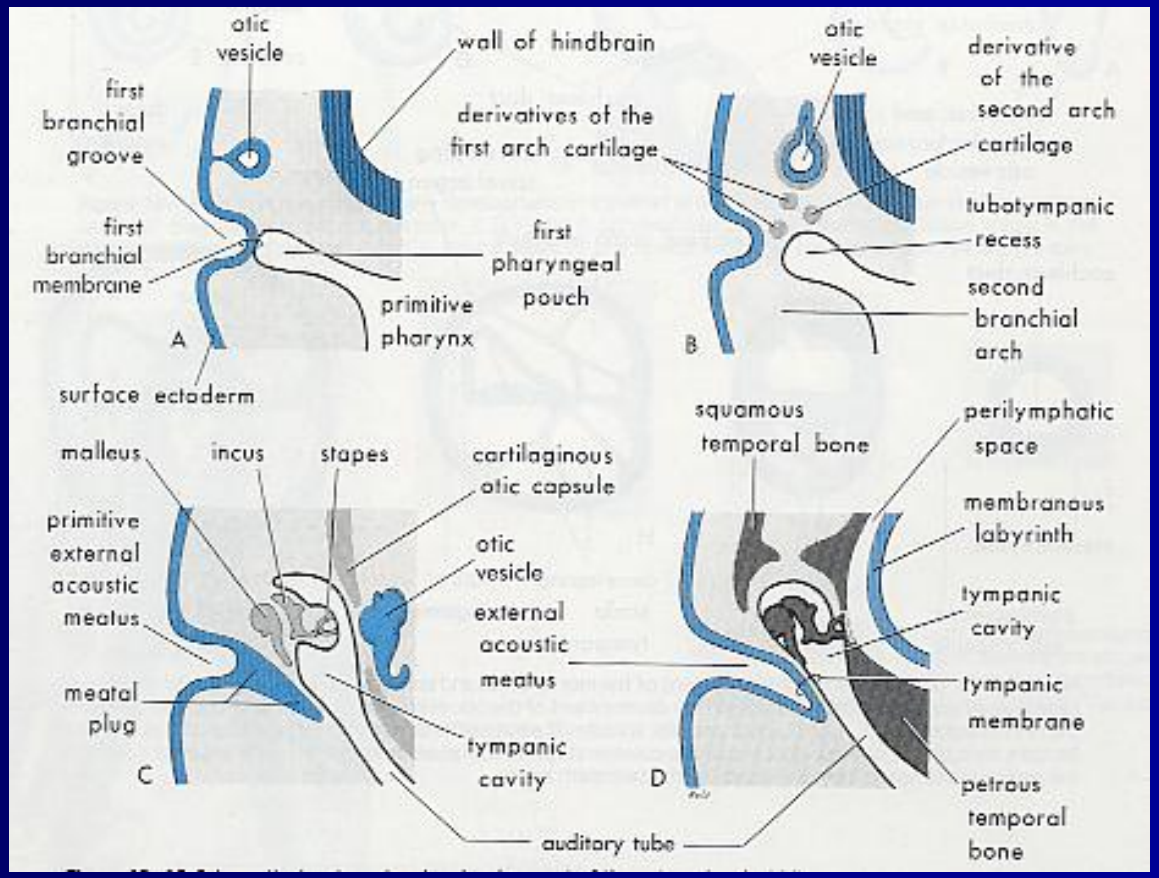
- Saccule and cochlear duct (becomes cochlea)

Inner Ear


otic placode
developing hindbrain
surface ectoderm
B
notochord
mesoderm
otic pit
otic placode
D
neural tube
otic vesicle
F
surface ectoderm
otic vesicle
G


endolymphatic duct
diverticulum
A1
developing posterior semicircular duct
B1
C1
D1
endolymphatic duct and sac
semicircular duct
ampulla
utricle
saccule
cochlea
E
absorption focus
saccule
A
B
C
D
cochlear duct
utricular and saccular portions of otic vesicle
developing spiral organ
bony labyrinth
ductus reuniens

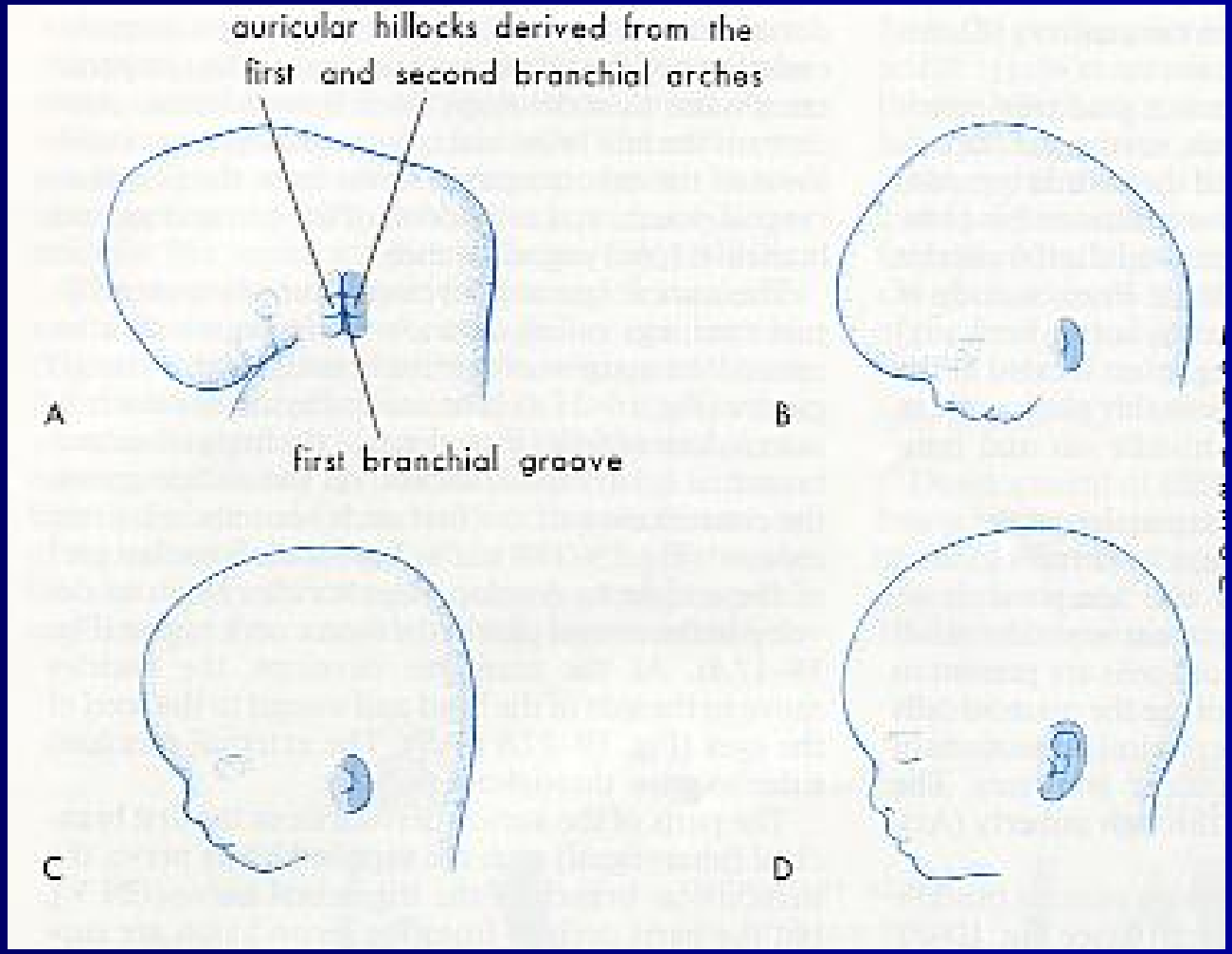
Middle Ear


otic vesicle
wall of hindbrain
first branchial groove
derivatives of the first arch cartilage
first branchial membrane
first pharyngeal pouch
primitive pharynx
A
surface ectoderm
otic vesicle
derivative of the second arch cartilage
tubotympanic recess
second branchial arch
B
malleus
incus
stapes
cartilaginous otic capsule
primitive external acoustic meatus
otic vesicle
external acoustic meatus
meatal plug
tympanic cavity
C
auditory tube
squamous temporal bone
perilymphatic space
membranous labyrinth
tympanic cavity
tympanic membrane
petrous temporal bone
D

External Ear

- Develops from 6 mesenchymal swellings, called **auricular hillocks**, which develop around the first branchial cleft
- The mesenchyme is derived from mesoderm in the first and 2nd branchial arches. As the ear grows, contrbutions of the first branchial arch become reduced.
- The pinna initially develops in the neck. As the mandible grows, it moves up to the level of the eyes.
- Part of the auricle originating from the first branchial arch is innervated by CN V
- The part originating from the 2nd arch is innervated by the cervical plexus (namely the lesser occipital and greater auricular nerves)

External Ear



External Ear

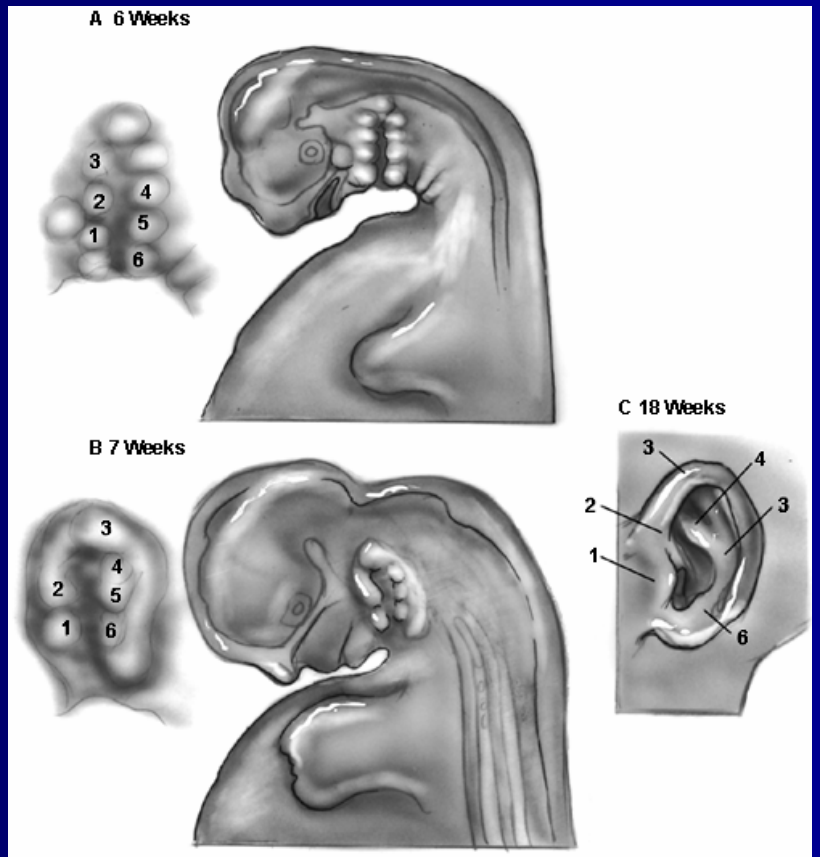


FIG. 5-1. Development of the auricle. A: Six hillocks form on the first and second branchial arches; all are identifiable at 6 weeks. B: Seven-week stage. C: By 18 weeks, the adult form is recognizable.

Thank You